

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T38 · Differentiation

Questions

Student book · questions with writing space

■ 18 pages

FAMILY MAP

Differentiation

F01	Find and classify stationary points	3 Q 17 marks
F02	Use the derivative as gradient or rate	6 Q 29 marks
F03	Apply differentiation to linear kinematics	4 Q 19 marks
F04	Optimise a practical model	3 Q 15 marks

17 $y = 4x^3 + 5x^2 + 2x$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots (2)$$

(b) Find the coordinates of the turning points on the graph with equation $y = 4x^3 + 5x^2 + 2x$
Show clear algebraic working.

.....
(4)

(Total for Question 17 is 6 marks)

Question 08

4MA1-1H Q18

MODULAR UNIT 2

5 marks

18 A curve C has equation $y = x^3 - 40x + 1$

Find the coordinates of both the points on C at which the gradient is 8

(..... ,)

(..... ,)

Question 13

4MA1-1H Q16 demand

MODULAR UNIT 2

4 marks

16 A curve has equation $y = 4x^3 - 8x + 5$

Find the x coordinates of the two points on the curve where the gradient is $\frac{1}{3}$

.....

4MA1-2H Q18 whole

4 marks

- $t =$

Question 11

4MA1-1H Q11

MODULAR UNIT 2

6 marks

11 The curve **C** has equation $y = 4x^3 + x^2 - 20x$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots$$

(2)

(b) Find the x coordinates of the points on **C** where the gradient is 4
Show clear algebraic working.

$$\dots\dots\dots$$

(4)

Question 13

4MA1-1H Q20 M01

MODULAR UNIT 2

4 marks

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.

Show clear algebraic working.

.....

Question 15

4MA1-1H Q17 Q17

MODULAR UNIT 2

5 marks

- 17 A particle P moves along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds, $t \geq 1$, is s metres where

$$s = 4t^2 + \frac{125}{t}$$

The velocity of P at time t seconds, $t \geq 1$, is v m/s

Work out the distance of P from O at the instant when $v = 0$

..... m

Question 15

4MA1-1H Q24

MODULAR UNIT 2

6 marks

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**

The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3

Show clear algebraic working.

$y = \dots\dots\dots$

Question 35

4MA1-2H Q24

MODULAR UNIT 2

5 marks

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geq 0$, is given by

$$x = 4t^3 - 27t + 8$$

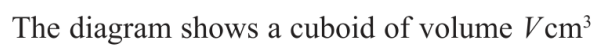
The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is $a \text{ m/s}^2$

Find the value of a

$a = \dots\dots\dots$

Diagram **NOT**
accurately drawn



- There is a value of x for which the volume of the cuboid is a maximum.

- (b) Find this value of x .
Show your working clearly.
Give your answer correct to 3 significant figures.

The centre of the circular face of the hemisphere and the centre of the top face of the cylinder are at the same point.

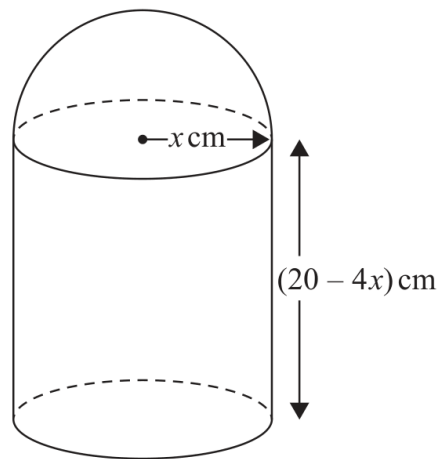


Diagram **NOT**
accurately drawn

The volume of **S** is $V\text{cm}^3$ where $V = \frac{1}{3}\pi y$

Find the maximum value of y .
Show clear algebraic working.

- At time t seconds where $t \geq 0$, the displacement, s metres, of P from O is given by

$$s = t^3 + 5t^2 - 8t + 10$$

Find the displacement of P from O when P is instantaneously at rest.

Give your answer in the form $\frac{a}{b}$ where a and b are integers.

..... metres

- Show your working clearly.

$$a = \dots\dots\dots$$

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x , in terms of p and m , for which the gradient of **C** is negative.

(Total for Question 23 is 4 marks)

The height of the cylinder is $\left(\frac{800}{\pi x} - x\right)$ cm.

Find the maximum value of V

Give your answer correct to the nearest whole number.

(Total for Question 20 is 5 marks)

Show clear algebraic working.